STAT 432 Homework 02

Conner Jensen (connerj2)

Contents

Question 1: Training and test error under a fixed design **1**

(a) . . . 2

(b) . . . 2

(c) . . . 3

Question 2: Prediction error at one target point **4**

(a) . . . 4

(b) . . . 4

(c) . . . 6

Question 3: The optimism correction **6**

(a) . . . 6

(b) . . . 6

(c) . . . 6

Question 4: Comparing Mallows' C_p, AIC, and BIC **7**

(a) . . . 8

(b) . . . 8

(c) . . . 8

Question 5: Validation and final test data **9**

(a) . . . 9

(b) . . . 9

(c) . . . 9

Reference **9**

Question 1: Training and test error under a fixed design

$n = 100$, $p = 20$, $\sigma^2 = 1$. One design $\mathbf{X}_{\text{all}}$ is drawn and then held fixed. $\beta_j = 0.4\sqrt{j}$ and $\boldsymbol{\mu} = \mathbf{X}_{\text{all}}\boldsymbol{\beta}$. Each of 200 runs draws independent $\mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\epsilon}$ and $\mathbf{y}^* = \boldsymbol{\mu} + \boldsymbol{\epsilon}^*$. For $m = 0, \ldots, p$, the fitted mean uses an intercept and the first m columns.

(a)

```
set.seed(43202)
n <- 100
p <- 20
nsim <- 200
X_all <- matrix(rnorm(n * p), n, p)
beta <- 0.4^sqrt(seq_len(p))
mu <- as.numeric(X_all %*% beta)

train_mat <- test_mat <- matrix(NA_real_, nsim, p + 1)
for (s in seq_len(nsim)) {
  y <- mu + rnorm(n)
  ystar <- mu + rnorm(n)
  for (m in 0:p) {
    X <- design(X_all, m)
    yhat <- as.numeric(X %*% qr.coef(qr(X), y))
    train_mat[s, m + 1] <- mean((y - yhat)^2)
    test_mat[s, m + 1] <- mean((ystar - yhat)^2)
  }
}

avg_train <- colMeans(train_mat)
avg_test <- colMeans(test_mat)
train_never_increases <- !any(apply(train_mat, 1, function(v) any(diff(v) > 1e-10)))
train_never_increases
```

```
## [1] TRUE
```

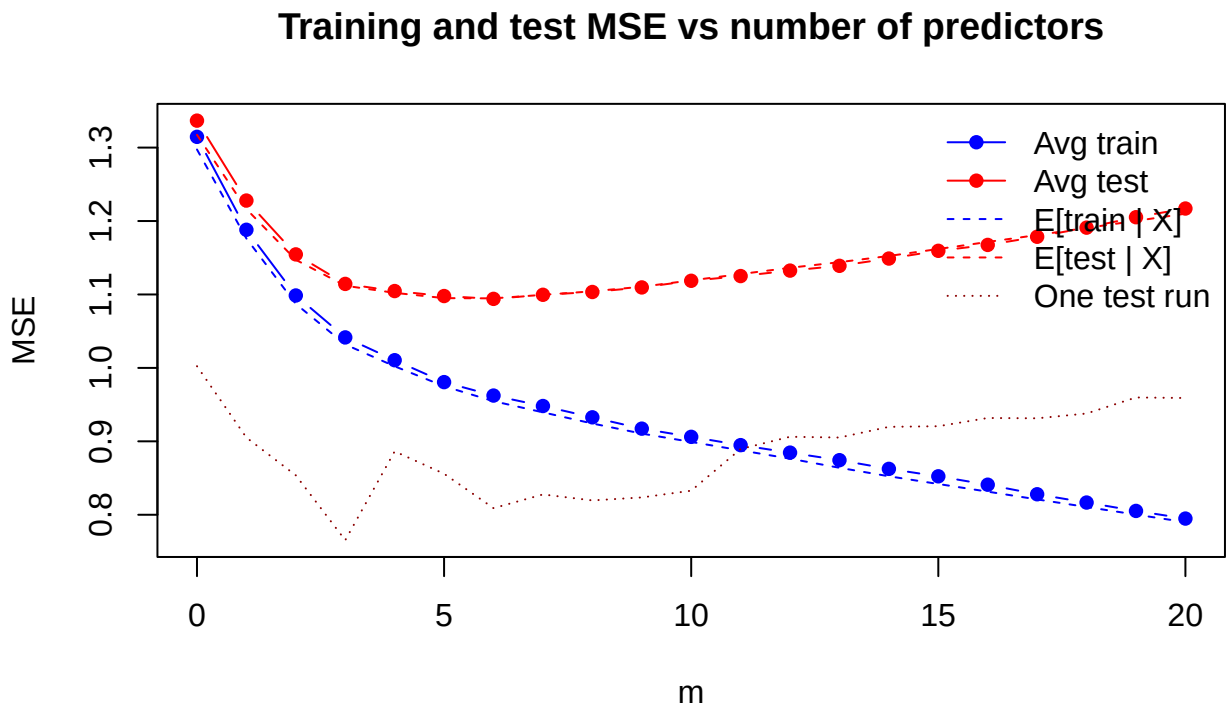
Training MSE did not increase when a predictor was added in any of the 200 runs. That is expected for nested OLS: adding a column cannot raise the in-sample residual sum of squares.

(b)

```
Bm2 <- sapply(0:p, function(m) {
  X <- design(X_all, m)
  mu_hat <- as.numeric(X %*% qr.coef(qr(X), mu))
  sum((mu - mu_hat)^2)
})
eth <- Bm2 / n + 1 - (0:p + 1) / n
ete <- Bm2 / n + 1 + (0:p + 1) / n
```

```
m_grid <- 0:p
plot(m_grid, avg_train, type = "b", pch = 16, col = "blue",
     ylim = range(avg_train, avg_test, eth, ete, test_mat[1, ]),
     xlab = expression(m), ylab = "MSE",
     main = "Training and test MSE vs number of predictors")
lines(m_grid, avg_test, type = "b", pch = 16, col = "red")
lines(m_grid, eth, lty = 2, col = "blue")
lines(m_grid, ete, lty = 2, col = "red")
```

```r
lines(m_grid, test_mat[1, ], lty = 3, col = "darkred")
legend("topright", bty = "n",
       legend = c("Avg train", "Avg test", "E[train | X]", "E[test | X]", "One test run"),
       col = c("blue", "red", "blue", "red", "darkred"),
       lty = c(1, 1, 2, 2, 3), pch = c(16, 16, NA, NA, NA))
```



The dashed curves are the conditional expectations given $\mathbf{X}_{\text{all}}$. They sit on top of the 200-run averages. Both average and expected test MSE bottom out at $m = 6$. Training MSE keeps falling all the way to $m = 20$.

(c)

The single-run test curve is jagged because it is one draw of $\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}^*$. Averaging 200 independent pairs removes that noise.

Write the expected test MSE as

$$\frac{B_m^2}{n} + 1 + \frac{m+1}{n}.$$

The first term is approximation bias and decreases as m grows. The last term is estimation variance and increases with m. Training MSE has the same bias but subtracts $(m+1)/n$ instead of adding it, so it is always pulled downward by extra coefficients. Selecting m by training MSE therefore favors the largest model and ignores the variance cost that shows up on new responses.

Question 2: Prediction error at one target point

```
n <- 100
p <- 6
sigma2 <- 1
X_all <- outer(1:n, 1:p, function(i, j) sqrt(2) * cos(pi * j * (i - 0.5) / n))
beta <- rep(0.5, p)
x0 <- c(0, 1, 0, 0, 1, 0)
mu0 <- as.numeric(crossprod(x0, beta))
mu0
```

```
## [1] 1
```

So $\mu_0 = \mathbf{x}_0^{\mathrm{T}}\boldsymbol{\beta} = 1$.

(a)

```
q2_theory <- t(sapply(0:p, function(m) {
  bias2 <- if (m == p) 0 else sum(x0[(m + 1):p] * beta[(m + 1):p])^2
  vterm <- (sigma2 / n) * (1 + if (m == 0) 0 else sum(x0[seq_len(m)]^2))
  c(m = m, squared_bias = bias2, variance = vterm, expected_se = bias2 + vterm)
}))
q2_theory
```

```
##      m squared_bias variance expected_se
## [1,] 0         1.00     0.01        1.01
## [2,] 1         1.00     0.01        1.01
## [3,] 2         0.25     0.02        0.27
## [4,] 3         0.25     0.02        0.27
## [5,] 4         0.25     0.02        0.27
## [6,] 5         0.00     0.03        0.03
## [7,] 6         0.00     0.03        0.03
```

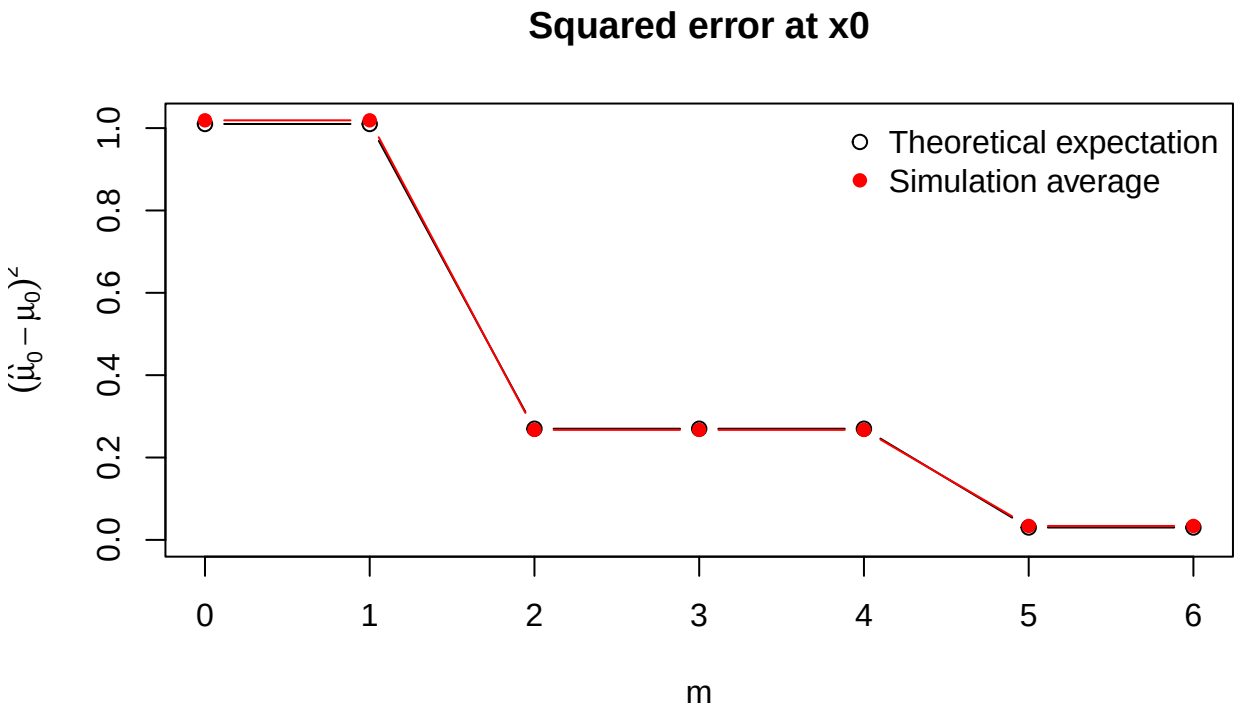
The first term is squared bias from dropping predictors that still enter μ_0. The second term is variance from estimating the intercept and the included coefficients; the $1/n$ is the intercept, and each included x_{0j}^2 adds a coefficient-variance contribution. Because this design is orthogonal, unused columns do not affect the included estimates.

Expected squared error is 1.01 at $m = 0, 1$; 0.27 at $m = 2, 3, 4$; and 0.03 at $m = 5, 6$. It changes only when predictor 2 or 5 is added. It is unchanged for $0 \to 1$, $2 \to 3 \to 4$, and $5 \to 6$.

(b)

```
set.seed(43203)
nsim <- 200
mu <- as.numeric(X_all %*% beta)
sqerr <- matrix(NA_real_, nsim, p + 1)
for (s in seq_len(nsim)) {
```

```r
  y <- mu + rnorm(n)
  for (m in 0:p) {
    X <- design(X_all, m)
    bhat <- qr.coef(qr(X), y)
    x0m <- c(1, if (m == 0) numeric(0) else x0[seq_len(m)])
    sqerr[s, m + 1] <- (sum(x0m * bhat) - mu0)^2
  }
}
sim_mean <- colMeans(sqerr)
plot(0:p, q2_theory[, "expected_se"], type = "b", pch = 1, ylim = range(0, q2_theory[, "expected_se"], s
     xlab = expression(m), ylab = expression((hat(mu)[0] - mu[0])^2),
     main = "Squared error at x0")
lines(0:p, sim_mean, type = "b", pch = 16, col = "red")
legend("topright", bty = "n",
       legend = c("Theoretical expectation", "Simulation average"),
       pch = c(1, 16), col = c("black", "red"))
```




```r
min_m <- min(which(q2_theory[, "expected_se"] == min(q2_theory[, "expected_se"]))) - 1
min_m
```

```
## [1] 5
```

The simulation averages track the formula. The smallest m that minimizes the theoretical error is $m = 5$.

(c)

$\mu_0 = \sum_j x_{0j}\beta_j$, and $\mathbf{x}_0$ is zero except in coordinates 2 and 5. Only those two predictors contribute to the mean at this target, even though every β_j is 0.5. Adding a predictor with $x_{0j} = 0$ does not change bias at $\mathbf{x}_0$ and, for this orthogonal design, does not change the variance term either.

Question 1's test MSE averages squared error over all n rows of $\mathbf{X}_{\text{all}}$. That average feels every omitted β_j. The target-specific error at $\mathbf{x}_0$ only cares about the coordinates that are nonzero there.

Question 3: The optimism correction

Let $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$. Then $\mathbf{M}^\mathsf{T} = \mathbf{M}$, $\mathbf{M}^2 = \mathbf{M}$, $\mathbf{MH} = \mathbf{0}$, and $\operatorname{tr}(\mathbf{M}) = n - (p+1)$.

(a)

$$\mathbf{y} - \mathbf{H}\mathbf{y} = \mathbf{M}\boldsymbol{\mu} + \mathbf{M}\boldsymbol{\epsilon}.$$

Expanding the squared length and taking the conditional expectation,

$$E(\|\mathbf{y} - \mathbf{H}\mathbf{y}\|_2^2 \mid \mathbf{X}) = B^2 + E(\boldsymbol{\epsilon}^\mathsf{T}\mathbf{M}\boldsymbol{\epsilon} \mid \mathbf{X}) = B^2 + \sigma^2 \operatorname{tr}(\mathbf{M}) = B^2 + \sigma^2\big(n - (p+1)\big).$$

Dividing by n gives

$$E(\text{MSE}_{\text{train}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2\Big(1 - \frac{p+1}{n}\Big).$$

For the test error,

$$\mathbf{y}^* - \mathbf{H}\mathbf{y} = \mathbf{M}\boldsymbol{\mu} + \boldsymbol{\epsilon}^* - \mathbf{H}\boldsymbol{\epsilon}.$$

Independence of $\boldsymbol{\epsilon}^*$ and $\boldsymbol{\epsilon}$, the mean-zero assumption, and $\mathbf{MH} = \mathbf{0}$ kill the cross terms. Then

$$E(\|\mathbf{y}^* - \mathbf{H}\mathbf{y}\|_2^2 \mid \mathbf{X}) = B^2 + E(\|\boldsymbol{\epsilon}^*\|_2^2) + E(\boldsymbol{\epsilon}^\mathsf{T}\mathbf{H}\boldsymbol{\epsilon}) = B^2 + n\sigma^2 + \sigma^2(p+1).$$

Dividing by n gives

$$E(\text{MSE}_{\text{test}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2\Big(1 + \frac{p+1}{n}\Big).$$

(b)

Subtracting the two expectations,

$$E(\text{MSE}_{\text{test}} - \text{MSE}_{\text{train}} \mid \mathbf{X}) = \frac{2\sigma^2(p+1)}{n}.$$

With $n = 120$, $p = 4$, $B^2/n = 0.04$, and $\sigma^2 = 1$,

$$E(\text{MSE}_{\text{train}} \mid \mathbf{X}) = 0.04 + 1 - \frac{5}{120} = 0.9983,$$

$$E(\text{MSE}_{\text{test}} \mid \mathbf{X}) = 0.04 + 1 + \frac{5}{120} = 1.0817,$$

and the expected optimism is $10/120 = 1/12 \approx 0.0833$. This is an expectation over $(\boldsymbol{\epsilon}, \boldsymbol{\epsilon}^*)$. A single pair of responses can reverse the order.

(c)

```r
n <- 120
p <- 4
rss <- 116.4
s2 <- 0.96
mse_hat <- (rss + 2 * (p + 1) * s2) / n
cp <- rss / s2 - n + 2 * (p + 1)
c(mse_hat = mse_hat, Cp = cp, check = s2 * (1 + cp / n))
```

```
## mse_hat      Cp   check
##    1.05   11.25    1.05
```

So $\widehat{\text{MSE}}_{\text{test}} = 1.05$ and $C_p = 11.25$. The identity

$$\widehat{\text{MSE}}_{\text{test}} = \widehat{\sigma}^2\left(1 + \frac{C_p}{n}\right)$$

holds. If n and $\widehat{\sigma}^2$ are the same for every candidate, C_p and the corrected test MSE are strictly increasing functions of each other, so they have the same minimizer.

Question 4: Comparing Mallows' C_p, AIC, and BIC

```r
diabetes <- read.csv("data/diabetes.csv")
train <- diabetes[1:370, ]
n <- nrow(train)
rss_of <- function(preds) {
  fit <- lm(reformulate(preds, response = "y"), data = train)
  sum(resid(fit)^2)
}
pred_A <- c("bmi", "bp", "s5", "sex", "s1", "s2", "s4")
pred_B <- c(pred_A, "s6")
pred_full <- setdiff(names(train), "y")
rss_A <- rss_of(pred_A)
rss_B <- rss_of(pred_B)
rss_full <- rss_of(pred_full)
df_full <- n - (10 + 1)
s2 <- rss_full / df_full
crit <- function(rss, p) {
  data.frame(
    p = p,
    RSS = rss,
    Cp = rss / s2 - n + 2 * (p + 1),
    AIC_red = n * log(rss / n) + 2 * (p + 1),
    BIC_red = n * log(rss / n) + (p + 1) * log(n)
  )
}
data.frame(
  model = c("A", "B", "full"),
  p = c(7, 8, 10),
  RSS = c(rss_A, rss_B, rss_full)
)
```

```
##   model  p     RSS
## 1     A  7 1098909
## 2     B  8 1089717
## 3  full 10 1089452
```

```
c(full_residual_df = df_full, sigma2_hat = s2)
```

```
## full_residual_df       sigma2_hat
##          359.000         3034.684
```

```
rbind(A = crit(rss_A, 7), B = crit(rss_B, 8))
```

```
##   p     RSS       Cp  AIC_red  BIC_red
## A 7 1098909 8.116253 2974.640 3005.948
## B 8 1089717 7.087434 2973.532 3008.754
```

```
c(RSS_drop = rss_A - rss_B,
  n_log_ratio = n * log(rss_A / rss_B),
  Cp_AIC_penalty = 2,
  BIC_penalty = log(n))
```

```
##       RSS_drop    n_log_ratio Cp_AIC_penalty    BIC_penalty
##    9191.508850       3.107775       2.000000       5.913503
```

(a)

Model A has $p = 7$ and RSS $\approx 1.099 \times 10^6$. Model B has $p = 8$ and RSS $\approx 1.090 \times 10^6$. The full model has $p = 10$, RSS $\approx 1.089 \times 10^6$, residual degrees of freedom 359, and $\widehat{\sigma}^2 \approx 3034.7$.

(b)

C_p is 8.12 for A and 7.09 for B, so C_p selects B. Reduced AIC is 2974.64 vs 2973.53, so AIC selects B. Reduced BIC is 3005.95 vs 3008.75, so BIC selects A. These numbers are not comparable across criteria.

(c)

Adding `s6` drops RSS by about 9192. Relative to $\widehat{\sigma}^2$, that is a C_p/AIC improvement of about 3.11, which beats the extra penalty of 2 but not BIC's $\log n \approx 5.91$. AIC and C_p therefore prefer B; BIC prefers the smaller model A.

A selected model is the minimizer of a criterion on this training sample. It is not a claim that those predictors generated the data. Several models can be close, the full model is itself only a working approximation, and a different split or criterion can pick a different set.

Question 5: Validation and final test data

(a)

The first misuse is step 3: choosing the model with the smallest final-test MSE. The test rows were not used to fit coefficients, but they were used to choose m. The reported minimum is then the smallest of eleven noisy estimates. That minimum is biased downward as an estimate of the prediction error of the selected rule on a new sample from the same process.

(b)

Split the 370 training rows into five folds. For each $m = 0, \ldots, 10$ and each fold, fit the nested model on the other four folds and record MSE on the held-out fold. Average the five fold MSEs to get a CV score for each m. Select the m with the smallest CV score. Refit that model on all 370 training rows. Only after m is locked should the 72 final-test rows be used, and they should be used once, to estimate prediction error.

(c)

The eleven test MSEs can still be reported as a description of this particular split, with a statement that the minimum was used for selection. They should not be treated as an honest estimate of future error for the selected model. A new, unused test sample (or a pre-declared nested validation scheme that never touched those 72 rows for selection) would be needed for a fresh final evaluation.

Reference

James, Witten, Hastie, Tibshirani, and Taylor, *An Introduction to Statistical Learning*, Chapters 3, 5, and 6.